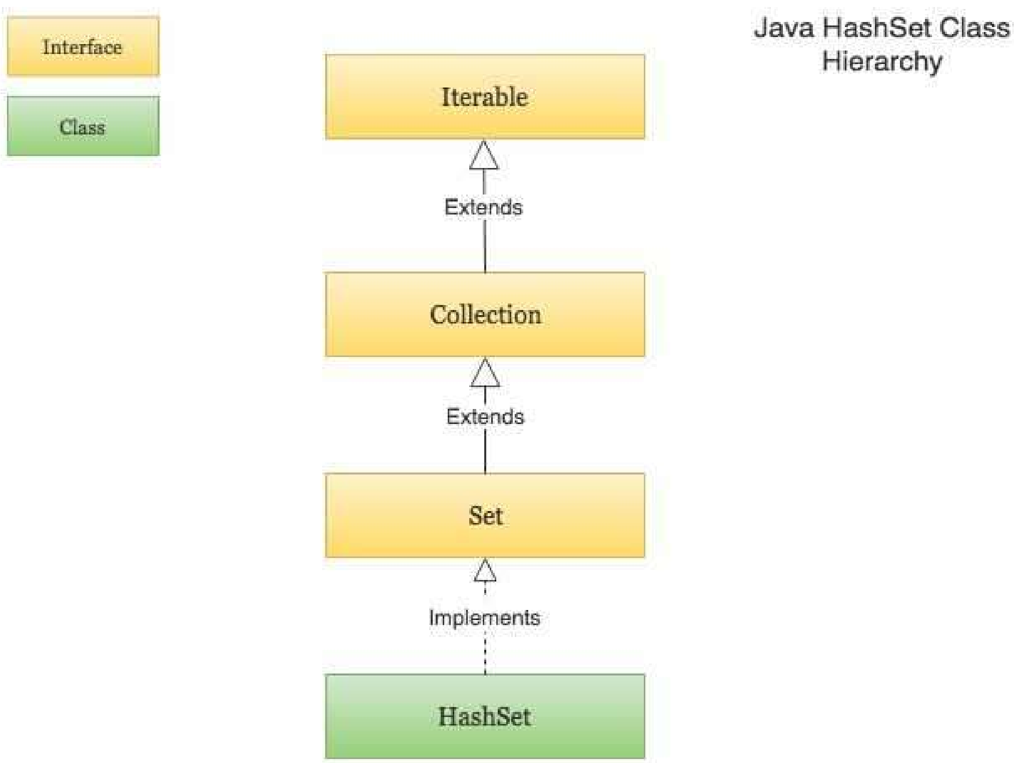
HashSet Interface in Collection

What is HashSet :-

- HashSet is a part of the **Java Collections Framework** and is used to **store unique elements.**
- It implements the Set interface and uses a **hash table** for storage.

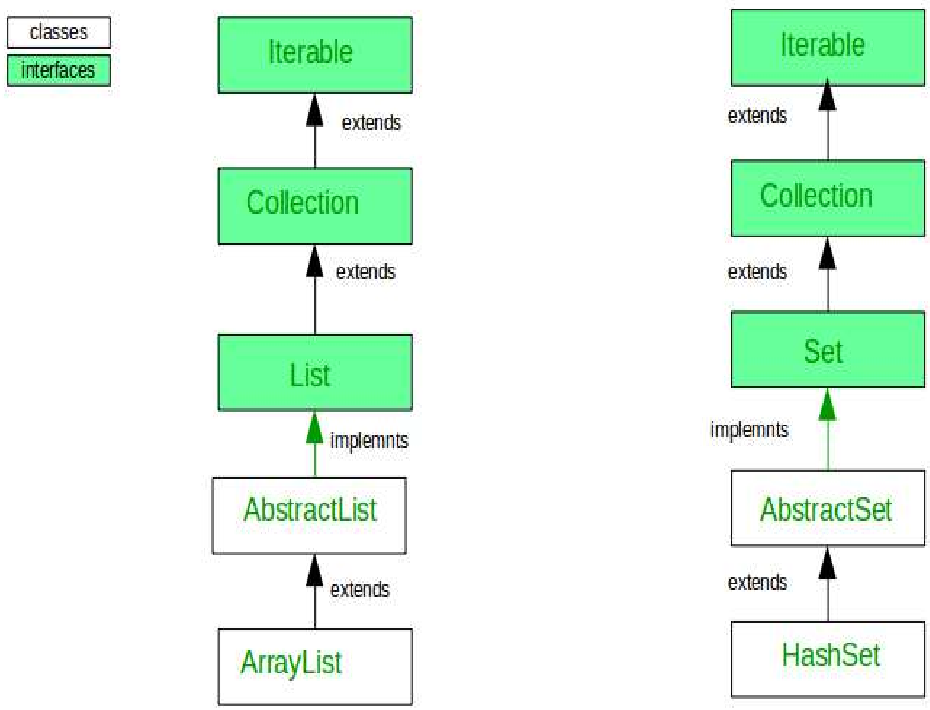


Key Points :-

- **Implements:** Set interface
- **Duplicates not allowed:** Automatically removes duplicate elements.

- **Order not maintained:** Does not maintain insertion order (unlike LinkedHashSet).
- **Null allowed:** At most one null element is allowed.
- **Fast operations:** Provides constant time performance for add, remove, and search (on average).
- **Backed by HashMap:** Internally uses a HashMap to store elements.

Difference between ArrayList and HashSet :-



- Use **ArrayList** → when you want ordered, indexed, or duplicate elements.
- Use **HashSet** → when you want **unique elements** and **don’t care about order**.

Methods in HashSet :-

Method	Description
add(E e)	Used to add the specified element if it is not present, if it is present then return false.
clear()	Used to remove all the elements from the set.
contains(Object o)	Used to return true if an element is present in a set.
remove(Object o)	Used to remove the element if it is present in set.
iterator()	Used to return an iterator over the element in the set.
isEmpty()	Used to check whether the set is empty or not. Returns true for empty and false for a non-empty condition for set.
size()	Used to return the size of the set.
clone()	Used to create a shallow copy of the set.

Implementation of Hashset :-

Input :-

```java
package Collection_Framework;

import java.util.HashSet;
public class hashset {

    public static void main(String[] args) {

        HashSet hs = new HashSet();
        hs.add("Nikhil");  // 1.add method
        hs.add(24);
        hs.add(24);
        hs.add(56.1);
        hs.add(null);
        hs.add(null);



        System.out.println("Hashset is : "+hs);
        System.out.println("Contains : "+hs.contains("Shinde")); // 2.contains method

        hs.remove(24);  // 3.remove object from collection
        System.out.println("After removing object hashset is : "+hs);

        System.out.println("is empty : "+hs.isEmpty()); // 4.isempty

        hs.clear(); // 5.clear
        System.out.println("Hashset is : "+hs);
    }
}
```

Output :-

```
Hashset is : [null, 56.1, Nikhil, 24]
Contains : false
After removing object hashset is : [null, 56.1, Nikhil]
is empty : false
Hashset is : []
```